2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.1. Habitat:

- This protozoon lives in freshwater ponds and streams that contain decaying vegetation.

2.1.2. Structure:

- It has a slipper-shaped body measuring about 0.25mm in length.
- It has a dorsal convex surface and a flat ventral surface.
- The anterior end is blunt, while the posterior end is pointed.
- The body is limited by a firm pellicle and the cytoplasm is differentiated into an outer clear area, the ectoplasm and an inner granular endoplasm.

2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.2. Structure (cont.):

- An almost uniform layer of small hair-like locomotory processes known as the cilia covers the body.
- On the flat surface of the animal there is a shallow groove, the oral groove that extends obliquely into the endoplasm to form a wide tube known as the vestibule that ends in a cytostome.
- The latter opens in a short narrow cytopharynx in which the cilia are arranged so close to each other as to form what looks like an undulating membrane.
- Just behind the cytopharynx there is a temporary anus or cytoproct.

2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.2. Structure (cont.):

- As in other ciliates, there are two dissimilar nuclei, which differ in size and appearance, a large kidney-shaped macronucleus and a small rounded micronucleus.
- The macronucleus appears to control the vegetative activities of the animal whereas the micronucleus is important for reproduction.
- The cytoplasm includes a number of food vacuoles as well as two contractile vacuole systems; one is located at the anterior end and the other is found near the posterior end of the body.
- Each contractile vacuole system is formed of a central vacuole into which leads a number of elongated pear-shaped radiating canals.
- The ectoplasm contains some highly differentiated organelles.
- When *Paramecium* is treated with special staining techniques, the pellicle shows a large number of hexagonal depressions, which are separated by ridges.

2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.2. Structure (cont.):

- From the center of each of these depressions projects a cilium that originates from a basal granule known as the kinetosome.
- The kinetosomes of each longitudinal row of cilia are joined together by longitudinal fibrils known as kinetodesmata.
- The kinetosomes of each transverse row of cilia are also joined by similar fibrils.
- Alternating between the bases of the cilia there are spindle-shaped structures known as the trichocysts, which open to the exterior by fine openings on the ridges between the hexagonal depressions.
- Upon irritation, these trichocysts are extruded to the exterior as long fine threads that act as defensive organelles.
- They may also act as means of attachment or anchorage of the animal to the substratum (Fig. 19).

2.2. Classification:
IV. SUBPHYLUM CILIOPHOR
2. Common species:
2.1. PARAMECIUM SP.:
2.1.2. Structure (cont.):

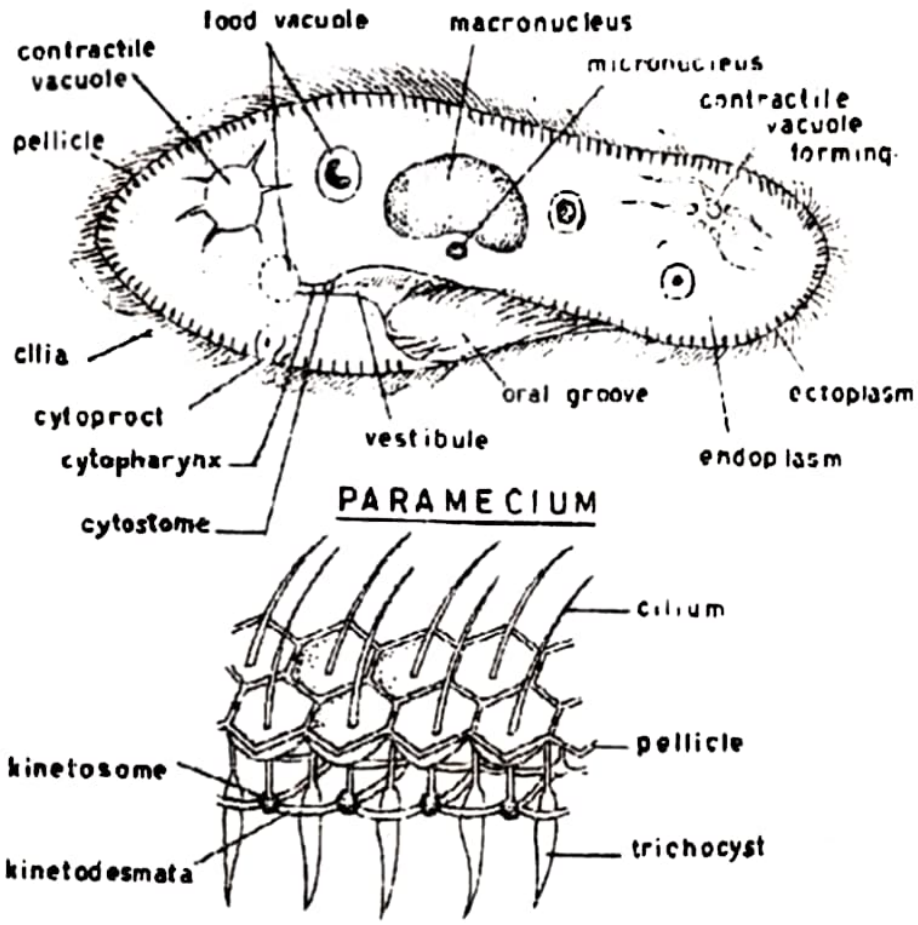

Fig. 19: Structure of *Paramecium sp.*

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.3. Locomotion:

- *Paramecium* swims rapidly in water by means of well-coordinated movements of numerous cilia found on the body of the animal.
- A cilium beats throughout its length and strikes the water so that the organism as whole tends to move in a direction opposite to that of the effective beat.
- Usually, a cilium on a single longitudinal row moves slightly in advance of the cilium behind it and slightly behind the one in front of it.

2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.3. Locomotion (cont.):

- Accordingly, the cilia on the same longitudinal row beat metachronously while those on the same transverse row are usually observed at the same phase of locomotion and thus are described as beating synchronously.
- The beating of cilia in this manner cannot be the only element in the locomotion of *Paramecium*, since the animal never moves in a straight line, but it follows an anti-clockwise (left) spiral path and at the same time it rotates along its longitudinal axis.
- There are three components, which apparently lead to this characteristic type of locomotion (Fig. 20).

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:

IV. SUBPHYLUM CILIOPHORA

2. Common species:

2.1. PARAMECIUM SP.:

2.1.3. Locomotion (cont.):

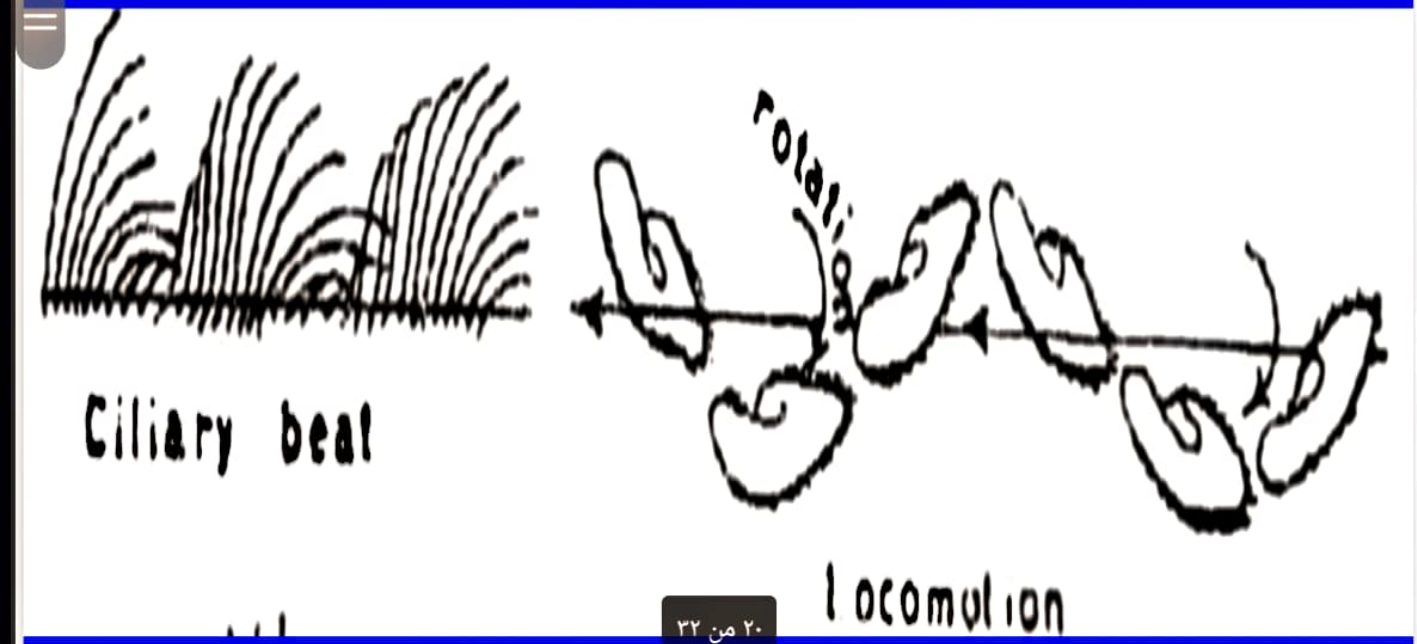

Fig. 20: Ciliary beat and locomotion in *Paramecium sp.*

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:

2.1.4. Reproduction:

- *Paramecium* reproduces both asexually (by transverse binary fission) and sexually (by conjugation and autogamy).

1-Transverse binary fission:

- Unlike many other protozoa, binary fission in *Paramecium* takes place in a transverse plane.
- The micronucleus divides by mitosis into two micronuclei that move towards the opposite ends of the body, and then the macronucleus divides transversely by mitosis.
- Other organelles such as the contractile vacuoles, oral groove, and cytopharynx become duplicated.
- Then a transverse furrow divides the cytoplasm into two almost equal parts.
- The resulting two daughter animals grow eventually to full size before another division is initiated.

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:

2.1.4. Reproduction:

1-Transverse binary fission (cont.):

- Transverse binary fission is completed within about 2 hours and occurs one to four times per day.
- However, this process cannot proceed indefinitely as the daughter paramecia become progressively less viable, some abnormalities appear in them, and the rate of their division gradually slows down until it eventually stops, as if they become senescent.
- This is because the macronucleus is highly polyploid and when it divides by mitosis; its chromosomes become divided at random between the two daughter paramecia.
- Repeated division of this sort naturally causes some disturbance in the chromosome structure of the resulting macronuclei, which results in a progressive decrease in the viability of the offspring. However, the animal resorts at intervals to a special type of sexual reproduction, known as conjugation, by which it regains its juvenescence, normal viability and power to reproduce by binary fission again.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:
1-Transverse binary fission (cont.):

- Transverse binary fission is completed within about 2 hours and occurs one to four times per day.
- However, this process cannot proceed indefinitely as the daughter paramecia become progressively less viable, some abnormalities appear in them, and the rate of their division gradually slows down until it eventually stops, as if they become senescent.
- This is because the macronucleus is highly polyploid and when it divides by mitosis; its chromosomes become divided at random between the two daughter paramecia.
- Repeated division of this sort naturally causes some disturbance in the chromosome structure of the resulting macronuclei, which results in a progressive decrease in the viability of the offspring. However, the animal resorts at intervals to a special type of sexual reproduction, known as conjugation, by which it regains its juvenescence, normal viability and power to reproduce by binary fission again.
- During that sexual process, the old macronuclei disintegrate and new ones with the original number of chromosomes are developed from the micronuclei, which are always diploid (Fig. 21).

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:
1-Transverse binary fission (cont.):

Fig. 21: Transverse binary fission in *Paramecium sp.*

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:

2- Conjugation:

- This process includes the following steps:
- a) Two individuals become sticky, adhere to one another by their oral surfaces and a protoplasmic bridge is formed between them.
- Each of these individuals is referred to as conjugant.
- b) The macronucleus in each conjugant breaks up into pieces and ultimately degenerates.
- In the meantime the micronucleus divides twice by mitosis to form four nuclei, each with the normal diploid number of chromosomes.
- Of these four nuclei three degenerate and only one is left.

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:

2- Conjugation (cont.):

- c) The remaining nucleus divides by a reduction division (meiosis) into two nuclei; a small migratory (male) nucleus and a larger stationary (female) nucleus.
- Each of these two nuclei is haploid, containing half the number of chromosomes found in the original nucleus.
- d) The migratory nucleus in each conjugant becomes active, move into the other conjugant and fuses with the stationary nucleus therein.
- Thus, in each conjugant a zygotic or fusion nucleus is formed which contains the original diploid number of chromosomes. When this fusion is completed, the two pairing paramecia separate and they are now referred to as the ex- conjugants.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:
2- Conjugation (cont.):

- e) The zygotic nucleus in each ex-conjugant divides thrice forming eight nuclei each with the diploid number of forming chromosomes.
- Four of these nuclei enlarge and become polyploid, thus forming macronuclei, while the other four remain as diploid micronuclei.
- This is followed by two cytoplasmic divisions and accordingly four new daughter paramecia are produced from every ex-conjugant, each with one macronucleus and one micronucleus.
- Worthy of notice is that the old macronuclei of the two conjugants disappear at the beginning of conjugation and new developed in the daughter individuals.
- Besides reorganizing the chromosomal structure of the macronuclei, conjugation involves the exchange of nuclear (hereditary) material between two individuals, which result in the production of better offspring.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:
3- Autogamy:

- This involves a similar process of nuclear re-organization as in conjugation, but occurs within one individual. It runs as follows:
- a) In one Paramecium the macronucleus disintegrates while the micronucleus divides twice forming 4 nuclei.
- b) Three of these nuclei disappear and the remaining one divides once more (by meiosis) into 2 haploid nuclei.
- c) These two nuclei fuse together to form a zygotic or fusion nucleus.
- d) The zygotic nucleus divides three successive times (by mitosis) forming 8 nuclei.
- e) The cytoplasm then undergoes two repeated divisions producing 4-daughterparamecia, each with two nuclei. One of these nuclei forms a macronucleus and the other a micronucleus.
- Autogamy is essentially similar to conjugation in that the macronucleus disintegrates at its beginning and becomes replaced by a new one derived from the products of division of the micronucleus. But while conjugation allows for exchange of hereditary characters between two individuals, autogamy gives pure lines (Fig. 22).

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.4. Reproduction:
3- Autogamy (cont.):

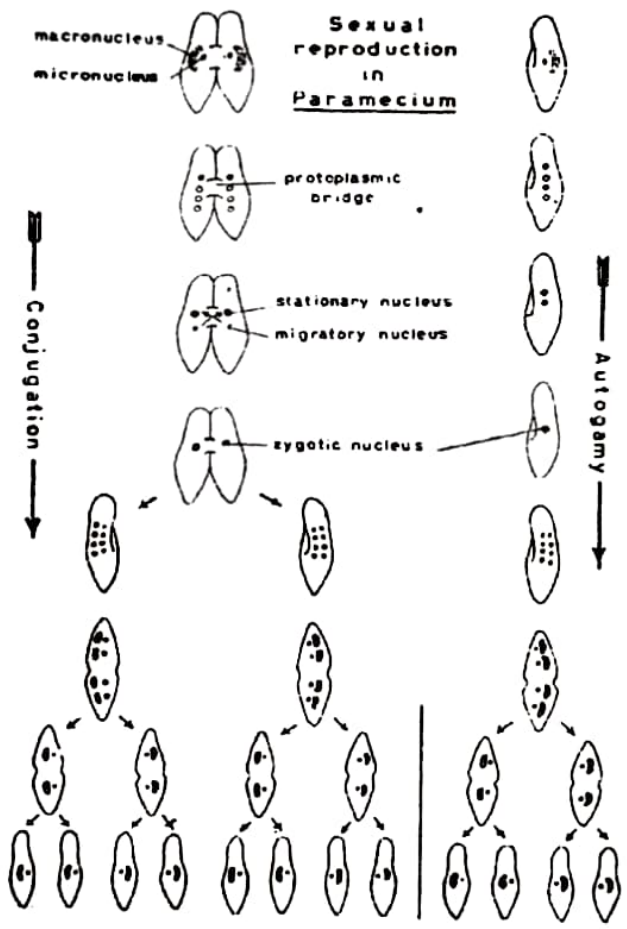

Fig. 22: Reproduction in *Paramecium sp.*

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.5. Sensitivity:

- *Paramecium* responds to different stimuli in much the same way like other free living protozoa described above.
- There are no specialized organs for sensation, although it is believed that the anterior end is more sensitive than the posterior one.
- One of the most interesting reactions of Paramecium is the avoiding reaction. If the animal is met with an adverse stimulus, it reacts negatively by moving away from that stimulus.
- Then it moves forwards and if it happens to face the same stimulus it will repeat the negative reaction, and by repeated trial and error it avoids that adverse stimulus.

2. SUBKINGDOM PROTOZOA (PHYLUM PROTOZOA)

2.2. Classification:
IV. SUBPHYLUM CILIOPHORA
2. Common species:
2.1. PARAMECIUM SP.:
2.1.5. Sensitivity (cont.):

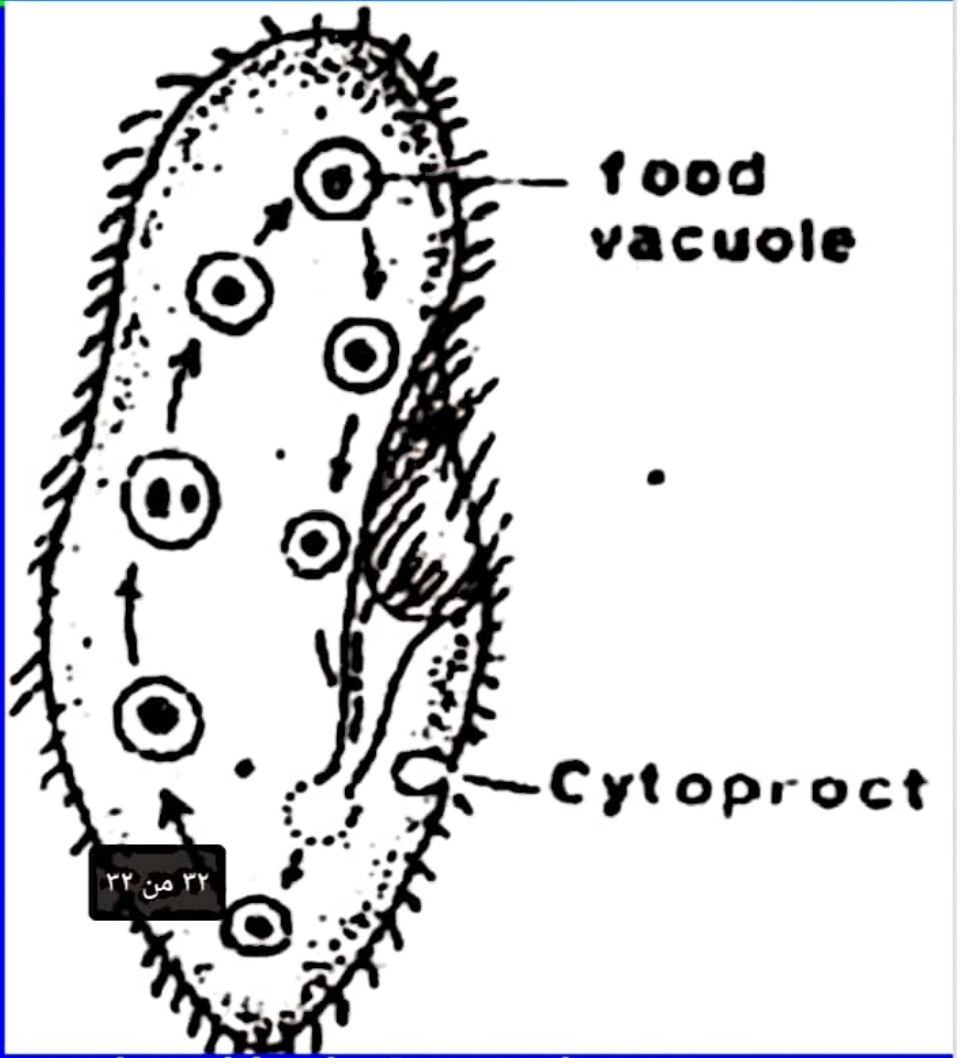

Fig. 23: Food capture and nutrition in *Paramecium sp.*